

Replication-only, real-time scenario-based questions for SQL Server Replication, grouped by key categories (Snapshot, Transactional, Merge, Peer-to-Peer, Filtering, Security, Performance & Tuning, Administration & Maintenance). This should give you deep coverage across all replication topics.

Categories & Question Counts (approximate)

Category	Focus	Approx # of Questions
Snapshot Replication	Initialization, snapshot agent, performance, locking, scheduling, cleanup	~ 60
Transactional Replication	Log reader, distribution agent, latency, initialization, schema changes	~ 100
Merge Replication	Conflict resolution, metadata cleanup, synchronization offline, resolver logic	~ 80
Peer-to-Peer Replication	Bi-directional changes, identity ranges, conflict avoidance, topology upgrades	~ 50
Filtering / Customization	Vertical / horizontal filters, parameterized filters, performance tradeoffs	~ 40
Security & Networking	Agent security, cross-domain, firewall, encryption, trust, permissions	~ 40
Performance & Tuning	Agent tuning, latency, batching, network optimizations, I/O, load balancing	~ 80
Administration & Maintenance	Backup/restore, distributor design, topology changes, reinitialization, DR	~ 50

Snapshot Replication (Questions ~1–60)

1. The snapshot agent is taking hours to create the snapshot. What diagnostics do you run and what optimizations do you apply?
2. The snapshot share is on a slow disk. How do you relocate or optimize the snapshot folder?
3. Snapshot generation is causing locks and blocking user queries — how to reduce that impact?
4. You must snapshot a very large table (hundreds of millions of rows). How do you break it into manageable chunks or schedule it?
5. The snapshot share is remote, and file transfer is failing intermittently. How do you improve reliability?
6. Snapshot agent fails with “file share unavailable” errors. What checks do you perform (permissions, network, share path)?
7. You need to compress snapshot files to save network bandwidth. How do you configure that?
8. Snapshot generation consumes too much tempdb / memory. How to mitigate resource usage?
9. Snapshot initializes a subscriber but fails halfway. How do you resume or restart?
10. You want to initialize multiple subscribers using one snapshot to save work. How do you do that?
11. Snapshot agent is blocked by long-running transactions. How to minimize blocking (use of “pre-snapshot interval,” reduce snapshot size)?
12. Snapshot replication for very large BLOB columns is slow. How to optimize (e.g. chunking, FILESTREAM, exclude columns)?
13. The snapshot file size grows unexpectedly large. Which objects or indexes might be causing that and how to troubleshoot?
14. You need to schedule snapshots outside business hours to reduce impact. How do you automate scheduling?
15. The snapshot share path is changed. How do you reconfigure the publication and agents safely?
16. Snapshot agent overwrites a custom script at subscriber. How to preserve subscriber customizations?
17. Snapshot cleanup is delayed, causing storage issues. How to manage retention / cleanup settings?
18. Using BCP for snapshot delivery — when is it applicable and how to configure?
19. Snapshot agent shows “snapshot in use by another process” errors. What causes that and how to resolve?

20. Snapshot agent profile settings: what tuning options exist (max command size, packet size, retry count)?
21. Snapshot replication with partitioned tables: how to snapshot per partition and apply selective partitions?
22. Snapshot replication in environments with schema changes: how to snapshot only changed schema vs full snapshot?
23. How do you avoid blocking the entire table while snapshot agent reads data?
24. Snapshot initialization via backup restore instead of file snapshot: pros, cons, procedure.
25. You have multiple publications using same snapshot share. How to organize shares, folder structure, avoid collisions?
26. Snapshot agent errors because of file access issues or permissions. What accounts, share permissions need configuration?
27. Log reader conflicts with snapshot agent locks. How do you schedule to avoid that?
28. Snapshot agent runs too slowly over WAN link. How do you optimize transfer (compression, network tuning)?
29. Snapshot agent generates schema and indexes — some indexes are not needed on subscriber. How to exclude them?
30. You need snapshot replication only for schema (no data). How to configure snapshot to exclude data?
31. Snapshot replication initialization of many subscribers in parallel — how to manage concurrency and resource usage?
32. You realize snapshot agent may overrun window. How to cancel or pause safely mid-process?
33. Snapshot agent is failing because of memory pressure or timeout. How to increase timeouts or resources?
34. Snapshot agent generates invalid script (due to custom objects). How do you correct and patch snapshot scripts?
35. You want to snapshot only data changes since last snapshot (incremental snapshot). Is that possible? How?
36. Snapshot replication for filtered articles (vertical or horizontal). How to snapshot only relevant subset?
37. Snapshot agent reports “command failure” on specific table. How to isolate which table, script it, fix it?
38. You must replicate very large tables but skip the snapshot entirely for new columns. How to add article with minimal snapshot overhead?
39. Snapshot agent’s timeout or retry settings too aggressive. How to adjust for unreliable networks?
40. Snapshot agent uses BCP vs bulkload – how to choose and configure?
41. If subscribers are geographically distributed, how to host multiple snapshot shares or local snapshot servers?
42. Snapshot agent occasionally leaves residual “__SnapshotFiles” folder. How to detect and cleanup leftovers?
43. Reporting on snapshot size over time to plan capacity. Which metrics or tables to monitor?
44. Snapshot share path cleanup fails. How to script cleanup or maintain retention automatically?
45. Snapshot file format or transfer corruptions: how to verify integrity?
46. The snapshot contains objects not needed at subscriber (e.g. indexes, triggers). How to exclude them?
47. The snapshot generation itself is overloaded (CPU / I/O). How to throttle or batch snapshot tasks?
48. Snapshot agent has to run on read-only publisher instance. How to configure agent appropriately?
49. Snapshot replication initialization fails due to collation mismatch. How to detect and resolve mismatch?
50. Snapshot agent must script extended properties, permissions; subscriber should replicate them. How to ensure they are included?
51. Subscriber character set or collation differences: snapshot script fails. How to detect and adapt?
52. Snapshot replication for temporal tables: how to snapshot base tables plus history, configure properly?
53. Snapshot for encrypted / Always Encrypted columns: how to snapshot encrypted data or decrypt?
54. Snapshot of compressed tables (row/page compression): how to deliver compressed data?
55. Snapshot agent performance: how to parallelize generation or streaming?

- 56. Snapshot replication with file tables or FILESTREAM: what special handling?
- 57. Snapshot agent failing because of file system quotas or permissions; diagnosing OS level issues.
- 58. Snapshot replication of filtered partitions with dynamic filters; how to generate snapshot only for active partitions?
- 59. Snapshot initialization via backup & restore + incremental snapshot; combining methods.
- 60. Snapshot agent resource settings: how to limit CPU / memory usage so it doesn't starve other workloads?

Transactional Replication (Questions ~61–160)

- 61. Log Reader Agent is falling behind — what tools and DMVs do you query to find cause?
- 62. "Undistributed commands" accumulate — what causes them and how to clear backlog?
- 63. Transaction log is not truncating because replication hasn't processed transactions. How to fix?
- 64. Latency grows during peak activity. Which agent settings (e.g. polling interval, commit batch size) to adjust?
- 65. Distribution Agent fails with "timeout" or "constraint violation" errors. How to debug?
- 66. Transactional replication initialization via backup rather than snapshot. How to do it?
- 67. Adding new articles to existing publication without reinitializing subscriptions. Procedure and risks.
- 68. Handling schema changes in published tables: adding / dropping columns, altering data types. Best practices.
- 69. Split a large transactional replication into multiple publications for scalability. How to plan?
- 70. Subscriber apply throughput is slow. How to optimize Distribution Agent settings or subscriber indexes?
- 71. If the distribution database is overloaded, how to scale it (remote distributor, multiple distributors)?
- 72. Transactional replication over WAN: how to tune packet size, agent retries, timeouts?
- 73. Isolation levels: how they affect replication consistency and blocking (e.g. using snapshot isolation on publisher)?
- 74. Conflicts with foreign keys at subscriber when applying deletes/updates. How to manage constraint order?
- 75. Transactional replication with queued updating subscribers: how it works, tradeoffs, latency.
- 76. The log reader fails because of an open transaction on publisher. How to detect and resolve open transaction issues?
- 77. Large DDL operations (e.g. index rebuilds) cause log spikes; replication must keep up. How to manage?
- 78. The publisher is in full recovery mode; log backups must be frequent so replication doesn't block. How to schedule?
- 79. Subscriber is behind and distribution agent is being overwhelmed. What strategies (batching, filtering)?
- 80. Publisher identity values being replicated: how to configure "NOT FOR REPLICATION" and identity range management?
- 81. Altering published stored procedures: when can you alter vs drop and re-add?
- 82. Changing primary key or key columns in published table. How to coordinate across publisher and subscriber?
- 83. How to detect and handle missing transactions on subscriber?
- 84. Replication agent jobs interfering with other SQL jobs (backup, index maintenance). How to schedule to avoid conflicts?
- 85. Agent threshold settings: when to alert if latency exceeds threshold.
- 86. How to monitor replication performance via performance counters (e.g. "SQLServer:Replication Dist. Cmds/sec", "Syncer Latency").
- 87. Polling interval too small: overhead. Poll too infrequently: latency. How to pick?
- 88. How to handle large batch updates on publisher — break into smaller transactions to avoid overwhelming replication.
- 89. Ensuring that the publisher's replication contributions (commands) are queued efficiently in distribution.

90. Transactional replication with multiple subscribers and potential inconsistent latency across them. How to detect and balance?
91. Subscriber performance slow because of poor indexing to support apply process. How to optimize indexes on subscriber?
92. Constraints or triggers at subscriber affecting apply throughput — how to disable or optimize them temporarily?
93. Schema object dependencies — when replicating one article whose parent objects change. How to ensure proper order?
94. Distribution Agent fails because of duplicate key violations at subscriber — what might cause that and how to fix?
95. You need to skip certain large transactions (e.g. administrative operations) from replication. How to filter or exclude?
96. Agent memory or CPU usage is high; how to assess and limit resource consumption?
97. Generating summary metrics on replication throughput, commands applied per second, latency trends.
98. Replication between SQL Server versions (e.g. publisher version 2014, subscriber 2019) — version compatibility issues.
99. The subscriber's schema is altered manually and now conflicts with subsequent replication. How to reconcile?
100. Using transactional + updatable subscriptions: how to handle conflicts or latency.
101. Distribution agent fails on schema change (e.g. column drop). How to coordinate deployments?
102. Monitoring log reader performance: DMV msdb.dbo.MSlogreader_history, distribution agent DMVs.
103. Detecting segmentation, splitting commands to optimize throughput.
104. Using publisher or distributor CPU to parallelize reading / applying.
105. Diagnosing blocking between distribution agent and subscriber transactions.
106. Snapshot agent interfering with transactional replication jobs. How to coordinate schedules?
107. Use of "immediate_sync" vs "delayed_sync" setting in publication and its effect on subscription initialization.
108. Reinitializing a subscription gracefully when schema changed or data corrupted.
109. Detecting and removing stale subscription metadata.
110. Using partitioned publications and routing articles to different distributors.
111. Scaling distribution: combining remote distributor, multiple agents, scheduled agent windows.
112. Optimizing replication for heavy INSERT workloads: batching, commit frequency.
113. Monitoring replication memory usage via DMVs or perfmon.
114. Ensuring that log backups do not remove log entries needed for replication.
115. Handling large truncations or deletes at publisher: breaking into chunks to avoid replication overload.
116. Rewriting schema changes so that they do not require full reinitialization.
117. Ensuring that replication agents resume after failover.
118. Configuring high availability for distributor server in mission-critical environments.
119. Handling subscriber latency spikes during peak load; auto-throttling or pausing agents.
120. Detecting and diagnosing applying "command set too large" errors; splitting commands.
121. Using multiple Distribution Agents in parallel for large volumes.
122. Assessing performance impact of indexes on published tables (too many indexes slows writes & replication).
123. How to handle cascade deletes / updates replication properly (ensuring referential integrity).
124. Replication initialization via backup with subsequent incremental snapshot.
125. Transactional replication with soft deletes (flag columns) vs actual deletes; how that affects replication.

126. Evaluating and tuning agent profiles (default, custom) for distribution agent.
127. Detecting and responding to latency thresholds being breached; alerting and scaling.
128. Using DBCC SHRINKFILE or not on distribution DB or publisher — implications for replication.
129. Diagnosing replication agent job failure due to disk / I/O errors.
130. Using multiple subscriptions to balance load (e.g. some subscribers partial).
131. Handling subscriber schema version drift over time and automation to detect mismatches.
132. Ensuring transaction log chain stays intact when replication is active.
133. Handling replication when a large DDL (e.g. drop index) is run on publisher.
134. Monitoring for “commands processed/sec” performance counter and evaluating bottlenecks.
135. How to plan replication capacity: expected transactions/sec, latency budget, agent throughput.
136. Ensuring replication is robust across server restarts or failovers.
137. Balancing replication job priorities vs other workload.
138. Handling replication initialization when data volumes are huge — combining backup and snapshot.
139. Using partition switching for fast data loading and integrating replication.
140. Adjusting agent maximum command size / packet size to optimal values.
141. Diagnosing latency that is constant base lag even in quiet hours — root cause.
142. Using BCP / bulk load for applying commands at subscriber for speed in initial load.
143. Splitting articles among multiple publications to isolate heavy traffic tables.
144. Ensuring that replication does not block log truncation (via log reader backlog).
145. Monitoring and mitigating aborts / retries by distribution agent.
146. Ensuring distribution agent does not violate constraints at subscriber — adjusting order of commands.
147. Verifying that the subscriber’s hardware/network is adequate to apply incoming commands.
148. Using asynchronous vs synchronous commit on subscriber side (if involved).
149. Handling replication across high latency links: pipelining, compression, batching.
150. Ensuring replication metadata (MSrepl_commands, MSrepl_transactions) do not overflow.
151. Diagnosing replication when some subscribers receive updates but others don’t.
152. Ensuring minimal downtime when reinitializing critical tables.
153. Verifying that publication scripts (sp_addarticle etc.) are version-controlled.
154. Automating subscription creation via scripts or PowerShell.
155. Replication to read-only secondary replicas (e.g. AG secondaries).
156. Monitoring DMVs such as distribution..MSdistribution_history for trends.
157. Handling distribution database corruption. Recovery steps.
158. Adjusting distribution cleanup and retention windows to suit volume.
159. Ensuring consistent performance after SQL version upgrades or patches on publisher / subscriber / distributor.
160. Diagnosing performance regressions after schema changes while replication is active.

Merge Replication (Questions ~161–240)

161. Merge replication is generating many conflicts. How do you detect, analyze, and reduce conflict rate?
162. Subscriber is offline long enough that many changes have to merge. How do you optimize large merge sessions?
163. Metadata cleanup overhead is high (MSmerge_contents, MSmerge_tombstone). How to prune metadata?

164. Merge agent performance is poor; how to tune parameters (batch size, polling interval, partition grouping)?
165. Resolver logic customizations: how to write / deploy a custom conflict resolver?
166. Merge replication initial snapshot is large. How to minimize initial snapshot size?
167. Merge replication schema changes: how to propagate schema changes safely?
168. Subscriber drops an object manually; merge synchronization fails. How to recover subscriber?
169. Conflict columns detection: which columns to include, avoid unnecessary conflicts.
170. Handling identity ranges: how to allocate ranges to minimize collisions?
171. Using filter and parameterized sync in merge replication: design tradeoffs and implementation.
172. Subscriber triggers or constraints causing merge apply to fail. How to adjust?
173. Monitoring the merge agent history, latency, number of commands.
174. Merge latency becomes large. How to prioritize synchronization windows?
175. Conflict resolution via “publisher wins” or “subscriber wins” or custom. When to use which?
176. Merge replication security: agent accounts, permissions.
177. Subscriber schema drift (extra columns). How merge handles extra subscriber columns?
178. Merge replication snapshot applying schema and data; failures due to schema mismatches. How to detect and correct?
179. You need periodic reinitialization for certain subscribers. How to script selective reinit?
180. Merge replication for remote devices with intermittent connectivity (e.g. mobile). How to optimize sync usage?
181. Validate data consistency between publisher and subscriber after merges via checksums or row counts.
182. Merge replication with filtering: horizontal filters, join filters. How to set up and maintain?
183. Merge replication performance tuning: server resource constraints, partitioning merge work.
184. Handling conflicts due to time zone differences or datetime offsets.
185. Merge replication in partitioned databases. Challenges and solutions.
186. Subscriber resolution logic requiring business rules; how to integrate custom logic.
187. Conflict metadata growth over time; schedule cleanup jobs.
188. Merge replication initial snapshot fails due to BLOB columns. How to adjust schema.
189. Merge replication synchronization fails mid-stream. How to resume or repair?
190. Using partition grouping in merge replication to optimize apply and detection.
191. Adjusting agent timeout, abort threshold, parameter settings.
192. Merge of deletes: tombstones accumulation. How to manage tombstone tables sizes.
193. Merge agent performance diagnostic via DMVs and performance counters.
194. Conflict resolution across cascades or complex relationships.
195. Merge replication over WAN: batching, compression, conflict resolution latency.
196. Subscriber changes schema locally (new column). How merge handles extra columns and whether they replicate back.
197. Merge replication of computed or persisted computed columns; restrictions and behavior.
198. Merge replication initialization via backup + incremental snapshot.
199. Upgrading merge replication topology (add or remove subscribers) in live system.
200. Using custom conflict resolvers or DDL resolvers; how to deploy and version them.
201. Merge replication with hierarchical subscriber models.
202. Merge replication with peer-to-peer topologies (rare but possible).

203. Merge replication on large databases — splitting tables into separate publications.
204. Ensuring merge agent does not run during high traffic times to reduce impact.
205. Handling merge agent memory / CPU usage for heavy workloads.
206. Merge replication forcing reinit globally; consequences and recovery.
207. Logging / tracing merge agent activities (Extended Events).
208. Merge replication and identity range conflicts under heavy insert load.
209. Merge synchronization conflicts due to collisions — detection and resolution.
210. Merge replication with parameterized filters: passing subscriber parameters at sync.
211. Merge replication snapshot of filtered rows only; handling filter changes over time.
212. Merge replication initial snapshot fails due to constraints or triggers on subscriber. How to disable/enable properly?
213. Merge replication performance under heavy delete operations (tombstone growth).
214. Merge replication security — encryption, agent accounts privileges.
215. Merge replication performance regressions after schema changes.
216. Monitoring long sync times and diagnosing bottlenecks.
217. Splitting large merge replication publications across multiple agents.
218. Merging large changesets; batching, partitioning, applying in windows.
219. Diagnosing memory or resource pressure in merge agent.
220. Handling schema drift at subscriber: extra columns, missing columns.
221. Merge replication with participants that have different hardware capabilities.
222. Upgrading merge publication from SQL version to another; compatibility issues.
223. Merge replication using “download-only” or “upload-only” subscribers.
224. Merge replication initial snapshot failing due to schema mismatch or missing permissions.
225. Scheduling merge synchronization to avoid business hours conflicts.
226. Monitoring merge metadata tables (MSmerge_contents, MSmerge_tombstone) sizes, growth, pruning.
227. Handling subscriber conflict store retention (history).
228. Merge replication performance in high-latency networks.
229. Conflict resolution of multi-column updates; managing precedence.
230. Merge replication with cascading deletes or complex FK relationships.
231. Merge replication agent error handling: retry logic, error threshold, logging.
232. Merging data with varying schemas or custom subscriber modifications.
233. Merge replication conflicts in offline mode merging when many deltas exist.
234. Dynamic filter changes in merge replication; handling filter schema updates without full reinit.
235. Merge replication agent profile tuning: perform cleanup intervals, commit batch size, polling intervals.
236. Monitoring sync durations per subscriber; scaling high-latency subscribers differently.
237. Ensuring that merge apply operations on subscriber have adequate indexing, minimal blocking.
238. Handling identity reseeding after merge conflicts or resynchronization.
239. Validating that conflict resolution logic matches business rules over time.
240. Merge replication split or partitioning to improve scalability.

Peer-to-Peer Replication (Questions ~241–290)

241. You need to set up peer-to-peer replication. How do you assign identity ranges to avoid conflicts?
242. How to handle schema changes across all peers in peer-to-peer topology safely?

243. One node becomes unavailable. How do you ensure continuity and consistency?
244. Peer-to-peer replication has been in place and you must upgrade schema (add column). How to coordinate?
245. Diagnosing conflicts in peer-to-peer replication (duplicate key violations).
246. Handling identity insert conflicts or range overlaps across nodes.
247. When to use peer-to-peer vs transactional + read replicas.
248. Peer-to-peer in conjunction with snapshot replication for initialization.
249. Monitoring latency, throughput, and divergence across peers.
250. Managing topology changes: adding or removing nodes.
251. Handling data divergence: detecting and reconciling differences between peers.
252. Ensuring consistent primary key / constraints across all nodes.
253. Failover strategies in peer-to-peer: which node becomes “primary” or how to manage writes in failover.
254. Handling conflicting DML performed on multiple peers simultaneously.
255. Upgrading versions or patching nodes in a peer network. How to do rolling upgrades safely.
256. Peer-to-peer replication initialization of new nodes — using backup / snapshot.
257. Handling large data volume replication across nodes with acceptable latency.
258. Diagnosing peer-to-peer performance bottlenecks (network, apply lag).
259. Conflict prevention strategies (e.g. partitioning data ranges, application logic).
260. Managing identity ranges dynamically if a node runs out of its allocated range.
261. Preventing “looping” of changes across peers inadvertently.
262. Handling DDL replication across nodes: limit complexity and ensure consistency.
263. Peer-to-peer and read scale-out: read operations on any node. How to ensure fresh data.
264. Detecting and repairing a node that went out-of-sync (reinitialization, conflict reconciliation).
265. Peer-to-peer with subscribers (i.e. combining peer + subscriber relationships).
266. Performance tuning peer node application throughput.
267. Handling network partitions (split brain) in peer replication.
268. Ensuring that triggers or extra logic on one peer doesn’t cause unexpected replication loops.
269. Monitoring peer replication agent history and consistency.
270. Limiting write operations on peers during heavy replication to avoid saturation.
271. Configuring peer nodes with symmetric topology (every node replicates to every other).
272. Scaling peer network to many nodes — planning and configuration.
273. Coordinating schema changes (e.g. new table) across all peer nodes atomically.
274. Handling identity collisions if someone bypassed replication (manual insert) on one node.
275. Ensuring primary keys, constraints, and indexes consistent across all peers.
276. Upgrading peer nodes one by one, ensuring replication continuity.
277. Diagnosing peer divergence (some nodes have rows missing).
278. Reinitializing faulty node carefully without disrupting other peers.
279. Ensuring proper replication path ordering to avoid cycles.
280. Performance impact of peer replication overhead on write latency.
281. Handling merging topology with peer and subscriber relationships.
282. Choosing appropriate conflict resolution or prevention strategies in peer replication.
283. Monitoring memory / agent load on peer nodes.
284. Using monitoring tools or DMVs specific to peer replication.

- 285. Ensuring that peer nodes don't replicate the same change back (loop).
- 286. Designing partitioned peer replication (each peer manages subset).
- 287. Handling node isolation (temporarily disconnecting a peer).
- 288. Testing failover and recovery in peer network.
- 289. Ensuring agent security and permissions aligned across all peers.
- 290. Scaling peer network with geographic distribution and balancing latency.

Filtering / Customization (Questions ~291–330)

- 291. You need row filtering (horizontal replication) so only specific rows go to subscriber. How design filter predicate?
- 292. Design vertical filtering (only certain columns) on article. What are the pros/cons?
- 293. Parameterized filtering (e.g. each subscriber gets rows where region = @Region). How to set up?
- 294. Changing or expanding a filter predicate over time without reinitialization. Is it possible?
- 295. Performance impact of filters: how to measure and mitigate?
- 296. Using stored procedure filters vs static filter clauses—differences and tradeoffs.
- 297. Multi-column filters or join filters across tables: how to design and deploy them?
- 298. Filter predicates that use functions — how that can degrade performance; how to rewrite.
- 299. Nested filter logic: combining multiple filters. How to manage complexity?
- 300. Filtered replication with partitioned data: combining filters and partitions.
- 301. Ensuring filter predicates are sargable and can use indexes.
- 302. Monitoring filter selectivity to ensure efficient replication.
- 303. Side effects of filters on referential integrity at subscriber.
- 304. Ensuring parameterized filters scale when many subscribers have different filter values.
- 305. Handling filter changes: adding or removing filter columns/predicates.
- 306. Filtered articles vs full articles: determining which to filter.
- 307. Complex filter conditions (e.g. business logic in filter). How to implement in replication?
- 308. Versioning filter definitions over time; applying consistent filter across multiple subscribers.
- 309. Handling subscribers whose filter predicate overlaps (duplicate row issues).
- 310. Testing filtering logic correctness (ensuring subscriber gets correct rows).
- 311. Filtered replication behavior during reinitialization — only relevant rows?
- 312. Use of dynamic filters (e.g. filter based on subscriber metadata).
- 313. Handling filter predicate changes that necessitate reinit; how to minimize impact.
- 314. Monitoring performance of filtering (extra CPU, row determination cost) on publisher.
- 315. Using filtered indexes to support replication filters.
- 316. Combining filtering with vertical filtering (both row and column filters).
- 317. Handling subscribers with different filter sets uniformly.
- 318. Using filter descriptors (e.g. partition filter mapping).
- 319. Deleting data at publisher that filter prevents deletion at subscriber; how to reconcile?
- 320. Ensuring that filter logic does not break referential integrity at subscriber side.
- 321. Handling filter syntax changes between SQL Server versions (compatibility).
- 322. Filtered replication agent parameter tuning differently than non-filtered replication.
- 323. Monitoring undistributed commands in filtered replication — possible skew.
- 324. Using snapshot for filtered articles — snapshot agent must respect filters.

- 325. Filtered replication with merge replication; particular complexities or restrictions.
- 326. Handling filter predicate changes when new articles are added.
- 327. Handling joins in filtering across multiple related tables.
- 328. Impact of filters on indexing and execution plan optimization.
- 329. Conflicts or missing data due to incorrect filter logic.
- 330. Documentation and version control for filters — ensuring filter definitions survive migrations.

Security & Networking (Questions ~331–370)

- 331. Replication agents (Snapshot, Log Reader, Distribution) require specific permissions. What minimal privileges are needed?
- 332. Agents communicating across domains or forests. How to configure security/trust/credentials?
- 333. Firewall / port configuration: which ports or endpoints must be open for replication to work?
- 334. Encryption of replication traffic: how to enable TLS / SSL for agent communication?
- 335. Securing snapshot shares: NTFS / share permissions; preventing unauthorized access.
- 336. Agents running under least privilege accounts; avoiding using sysadmin where not needed.
- 337. Securing distributor, publisher, subscriber credentials; storing passwords safely.
- 338. Handling replication in environments where Windows / SQL authentication boundaries exist.
- 339. Auditing replication agent activity for security compliance.
- 340. Preventing unauthorized schema changes on replication objects.
- 341. Handling impersonation and execution context (ownership chaining) when replication applies DDL or DML.
- 342. Replication across secured network segments / DMZ; how to traverse securely.
- 343. Replication in multitenant / shared environments — ensuring isolation, permissions.
- 344. Encrypting snapshot files at rest; ensuring they cannot be tampered with.
- 345. Using certificates or Windows domain trust for agent authentication.
- 346. Securing replication metadata tables (MSrepl_*, distribution tables) from unauthorized access.
- 347. Replication integrity: detecting or preventing tampering of replicated data.
- 348. Handling replication across regimens with different collation / case sensitivity — security implications.
- 349. Managing and rotating replication agent passwords securely.
- 350. Logging agent failures and suspicious replication actions; integrating with security logs.
- 351. Replication in Always Encrypted environments; handling encryption at subscriber/publisher.
- 352. Auditing failed replication commands due to permission denial.
- 353. Proper use of “NOT FOR REPLICATION” to prevent replication paths triggering triggers or constraints inadvertently.
- 354. Replication security in cloud environments (Azure SQL, VMs): VPN, endpoints, access control.
- 355. Secure initialization (snapshot) files — signed scripts, integrity checks.
- 356. SQL Server service accounts and replication agent accounts isolation.
- 357. Preventing elevated privilege escalation via replication scripts.
- 358. Replication cross-server authentication (linked server style) if needed.
- 359. Protecting replication metadata and history from unauthorized deletion or tampering.
- 360. Configuring replication for least exposure (open only necessary endpoints).
- 361. Using secure credentials (avoid clear text passwords in scripts) for agent definitions.
- 362. Ensuring that subscriber customizations cannot break replication by manipulating replication artifacts.

- 363. Protecting replication from denial-of-service (agent spamming, snapshot floods).
- 364. Ensuring that replication agent logs sensitive data is masked or not exposed.
- 365. Verifying that replication initialization scripts (schema/data) do not expose sensitive table definitions inadvertently.
- 366. Securing replication topology control operations (adding/removing subscriber) to authorized DBAs only.
- 367. Using proxy accounts or certificate authentication in restricted environments.
- 368. Handling replication on Always On Availability Groups: security for read-only replicas.
- 369. Ensuring that replication initialization or distribution doesn't violate compliance (e.g. GDPR) by transferring restricted columns.
- 370. Security considerations when replicating across geographical or regulatory jurisdictions.

Performance & Tuning (Questions ~371–450)

- 371. Replication latency is consistently high. How do you profile and reduce latency?
- 372. Distribution agent is CPU bound — how to tune or offload work?
- 373. Log reader agent is I/O bound — how to identify bottlenecks and optimize?
- 374. Network bandwidth constraints are limiting replication throughput — what strategies to mitigate (compression, packet size tuning)?
- 375. Batching commands in replication: how to choose optimal batch size?
- 376. Monitoring replication performance over time to detect regressions.
- 377. Agent polling interval too frequent — overhead; too infrequent — latency. How to pick balance?
- 378. Distribution agent's commit threshold or max commands per batch tuning.
- 379. Applying commands too slowly at subscriber. How to speed up apply (indexing, disable constraints)?
- 380. Diagnosing replication overhead impact on publisher write performance.
- 381. Using multiple distribution agents in parallel for load splitting.
- 382. Splitting heavy articles among separate publications to isolate stress.
- 383. Evaluating whether distributing agents across multiple servers helps.
- 384. Using remote distributor to reduce load on publisher.
- 385. Monitoring undistributed/undelivered commands to ensure backlog doesn't grow.
- 386. Ensuring agent memory / buffers are sized properly.
- 387. Optimizing command shipping by grouping DML statements in fewer commands.
- 388. Monitoring performance counters specific to replication (e.g. "Articles Delivered/sec", "Commands Executed/sec").
- 389. Dealing with high write load and replication overhead; balancing tradeoffs.
- 390. Identifying bottlenecks in apply: blocking, locking, slow pages.
- 391. Using DMVs to find replication hotspots (e.g. sys.dm_repl_traninfo, msdb MSlogreader).
- 392. Detecting skew in subscriber lag across multiple subscribers — load balancing.
- 393. Agent job scheduling — spreading replication jobs to avoid overlapping heavy workloads.
- 394. Replication profile tuning (default profiles vs custom).
- 395. Slow subscribers causing cascading latency. How to isolate or throttle slower ones.
- 396. Agent error retry behavior — balancing between aggressiveness and resource consumption.
- 397. Monitoring and limiting resource usage (CPU / memory) of replication agents.
- 398. Diagnosing performance regressions after schema or index changes on published tables.
- 399. Using optimal indexing on published tables: too many indexes hurt write / replication performance.

400. Tuning distribution database itself: file placement, memory, index strategies.
401. Diagnosing slow startup of agents or long agent initialization.
402. Using trace flags for tuning replication (if supported) and careful use.
403. Using monitoring / logging to detect pattern of replication slowdowns (e.g. nightly spikes).
404. Determining whether latency is due to network, CPU, I/O, or subscriber slowness.
405. Replication agent memory pressure, internal buffer sizes; adjusting config.
406. Splitting replication agents to parallelize per-article or per-subscriber workloads.
407. Using replication throttling — limiting agent activity during peak times.
408. Diagnosing performance impact on publisher caused by snapshot, log reader, distribution loads.
409. Ensuring that agent jobs don't contend with backups or index maintenance.
410. Diagnosing apply performance using extended events / DMVs on subscriber.
411. Tuning agent retry counts, delay intervals.
412. Measuring per-article performance (which tables replicate slowly).
413. Using multiple data files or filegroups in distribution DB for performance.
414. Caching and prefetching in agent logic, if supported by SQL version.
415. Diagnosing skew in latency across subscribers and adjusting priorities.
416. Managing distribution retention / cleanup so distribution DB doesn't slow down over time.
417. Evaluating whether replication throughput meets SLA (e.g. commands/sec).
418. Detecting bottlenecks at snapshot or initialization time vs steady state replication.
419. Using derived tables or reduced article size to optimize replication.
420. Testing replication under load (stress testing) and measuring thresholds.
421. Diagnosing overhead of triggers, constraints, indexes on replication performance.
422. Optimizing replication agent disk access (logging, staging tables).
423. Ensuring distribution agent staging / temp tables are sized properly.
424. Diagnosing distribution DB I/O bottlenecks; moving DB files to faster disks.
425. Evaluating replication overhead on CPU when encryption or compression in use.
426. Using proper agent retry / backoff strategies to avoid overloading system under errors.
427. Diagnosing and resolving skew in replication latency across multiple nodes.
428. Agent memory / buffer tuning when replication involves high volume of change.
429. Using monitoring thresholds and autoscaling replication resources (e.g. more CPU on distributor).
430. Diagnosing uneven load among replication agents; rebalancing assignments.
431. Tuning parameterized agents (e.g. custom filters) for performance.
432. Diagnosing and preventing "thundering herd" when many agents start simultaneously.
433. Adjusting replication schedules (e.g. stagger agent start times).
434. Monitoring replication overhead on I/O subsystems and designing layout accordingly.
435. Using performance baselines to detect regression in replication performance.
436. Diagnosing latency variation across different articles (tables).
437. Tuning packet size / command size per replication agent.
438. Diagnosing replication effect on buffer pool / memory pressure.
439. Using query store or plan forcing for replication-related queries if needed.
440. Ensuring replication agents aren't swapping or using virtual memory under load.
441. Diagnosing and mitigating high latency caused by lock contention during apply.
442. Monitoring unseen slowdowns (background drift) in replication performance.

- 443. Tuning scheduler priorities or SQL Server priorities for replication agent processes.
- 444. Diagnosing skew in throughput between publisher and subscriber enviros.
- 445. Ensuring distribution DB statistics / indexes are maintained for replication usage.
- 446. Diagnosing agent caching or spool issues — when large command sets get blocked.
- 447. Throttling downstream subscribers so they don't overload publisher or distributor.
- 448. Diagnosing replication scheduling overlap with heavy workloads (e.g. report jobs).
- 449. Diagnosing replication in multi-tenant environments under heavy concurrency.
- 450. Optimizing replication for mixed workload (OLTP + reporting) simultaneously.

Administration & Maintenance (Questions ~451–500)

- 451. You must reconfigure replication topology (move distributor to new server). What steps to migrate?
- 452. The distributor server failed; how to recover distribution database and restore replication?
- 453. Reinitializing a subscription selectively (only certain articles) — how to script and perform safely.
- 454. Upgrading SQL Server version on publisher / subscriber without breaking replication.
- 455. Replication during disaster recovery: fallback subscriber promotion, reinitialization, syncing.
- 456. Scripting replication topology creation (publications, articles, subscriptions) for disaster recovery deployment.
- 457. Removing replication gracefully from a database: cleanup metadata, agent jobs, objects.
- 458. Monitoring replication agent job history retention and cleanup strategy.
- 459. Automating replication health checks and alerting via scripts or monitoring tools.
- 460. Changing or relocating distribution database (move to different disk or server).
- 461. Managing subscriber reinitializations when only part of data is stale.
- 462. Using “disable replication” option temporarily for maintenance and re-enable later.
- 463. Testing replication failover scenarios (publisher, distributor, subscriber).
- 464. Version controlling replication objects (scripts) for repeatability.
- 465. Handling replication metadata cleanup (distribution cleanup jobs, merge metadata).
- 466. Partitioning replication load across multiple distributors or remote distributors.
- 467. Performing offline schema changes with minimal replication disruption.
- 468. Automating subscription addition/removal via T-SQL or PowerShell.
- 469. Auditing replication changes (who added/removed subscriber, articles).
- 470. Documenting replication topology, filters, agent settings as part of DB governance.
- 471. Periodically verifying data consistency between publisher/subscriber (checksums, row counts).
- 472. Backup & restore of published databases: what precautions to take so replication remains intact.
- 473. Backing up the distribution database and its recovery steps.
- 474. Testing replication after restoration: broker artifacts, agent reconfiguration.
- 475. Handling filtered replication reinitialization after filter changes.
- 476. Changing replication from push to pull (or vice versa) on a subscription.
- 477. Performing replication cleanup (removing tombstone or metadata) without manual errors.
- 478. Migrating replication objects during server or DB rename/restore.
- 479. Scaling subscriber count: planning before adding hundreds of subscribers.
- 480. Upgrading to a new topology (e.g. from transactional to merge or adding peer).
- 481. Performing replication health audits: checking latency, errors, jobs, topology.
- 482. Automating reinitialization or repair of broken subscriptions with minimal manual intervention.

483. Temporarily stopping replication (e.g. for maintenance), capturing downstream changes, then resuming replication.
484. Documenting best practices, standard operating procedures for replication maintenance.
485. Delegating permissions to junior DBAs for replication tasks: what minimal rights.
486. Archiving or purging old subscription history / replication logs.
487. Monitoring disk usage for snapshot, distribution, log retention.
488. Testing replication upgrades across SQL version or edition changes (Standard → Enterprise) safely.
489. Disaster recovery planning: replication reestablishment, disaster scripts.
490. Re-initializing large subscribers using backup restore + incremental snapshot to minimize down time.
491. Handling replication in cloud migrations (on-prem → Azure VMs or Azure SQL).
492. Changing publication settings (immediate_sync, allowed schema changes) mid-lifecycle.
493. Merging snapshots from multiple sources (for multi-publisher scenarios).
494. Validating replication performance after hardware upgrade (IO, CPU, network).
495. Auditing replication agent account usage and credential rotation.
496. Handling reconfiguration of subscriber server names, database rename in replication.
497. Handling replication in multi-site DR setup: extended topology, failover paths.
498. Maintaining version control of replication scripts over schema versions.
499. Performing replication cleanup and reset in case of catastrophic failure (drop & rebuild).
500. Ensuring performance baselines and continual monitoring to detect degradation post maintenance or changes.